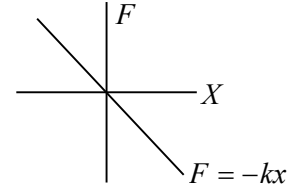


Simple Harmonic Motion

Any periodic motion is identified as simple harmonic motion if motion is periodic and there restoring force acting on a particle of mass m . If restoring force is linear with displacement from equilibrium then motion is identified as Simple harmonic motion. The best example of simple harmonic motion is spring mass system which is also identified as simple harmonic oscillator. The force corresponds to harmonic oscillator is shown in figure.



Equation of motion of simple harmonic oscillator $m \frac{d^2x}{dt^2} = -kx$

The equation reduces to $\frac{d^2x}{dt^2} + \omega^2 x = 0$ where $\omega = \sqrt{\frac{k}{m}}$ is identified as angular frequency of system for harmonic oscillator the angular frequency ω for oscillation is constant.

The solution of differential equation is $\frac{d^2x}{dt^2} + \omega^2 x = 0$ and $x = c_1 \cos \omega t + c_2 \sin \omega t$ the c_1 and c_2 can be determined by initial condition on position x and velocity $\dot{x} = \frac{dx}{dt}$.

The position $x = c_1 \cos \omega t + c_2 \sin \omega t$ can be also written as $x = A \cos(\omega t + \phi)$ where $c_1 = A \cos \phi$ and $c_2 = -A \sin \phi$.

The term A is identified as amplitude which is identified as maximum displacement and ϕ is initial phase. Initial phase will

$$A = \sqrt{c_1^2 + c_2^2} \text{ and } \phi = \tan^{-1} \left(-\frac{c_2}{c_1} \right)$$

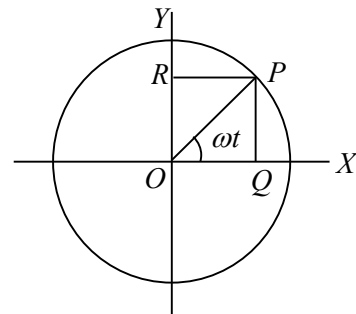
The velocity can be given by $\dot{x} = \frac{dx}{dt} = -A\omega \sin(\omega t + \phi)$

But best solution of the $\frac{d^2x}{dt^2} + \omega^2 x = 0$ x is equal to Real part of function $A \exp i(\omega t + \phi)$.

We can write $x = \text{Re} [A \cos(\omega t + \phi) + i f(t)]$

If we choose the simplest $f(t)$ as $f(t) = A \sin(\omega t + \phi)$ then the locus between real part $x = A \cos(\omega t + \phi)$ and $y = A \sin(\omega t + \phi)$

The solution of simple harmonic oscillator is identified as projection of radius vector with magnitude A along x or real axis.



Energy of Harmonic Oscillator

The kinetic energy of system is given by $T = \frac{1}{2}m\dot{x}^2$ The potential energy of system is given by

$$V = -\int F dx = -\int -kx dx = \frac{1}{2}kx^2 + c \text{ we have initial condition as } x=0, V=0 \text{ so } c=0$$

So potential energy is given by $V = \frac{1}{2}kx^2$

Total energy $E = T + V = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$ where $k = m\omega^2$ so $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\omega^2 x^2$

$x = A \cos(\omega t + \delta)$ then the velocity is given by $\dot{x} = -A\omega \sin(\omega t + \delta)$

Total energy is given by $E = \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t + \delta) + \frac{1}{2}m\omega^2 A^2 \cos^2(\omega t + \delta) = \frac{1}{2}m\omega^2 A^2$

We can see total energy of harmonic oscillator is constant

So, one can write $\frac{1}{2}m\omega^2 A^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\omega^2 x^2 \Rightarrow \frac{1}{2}kA^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$

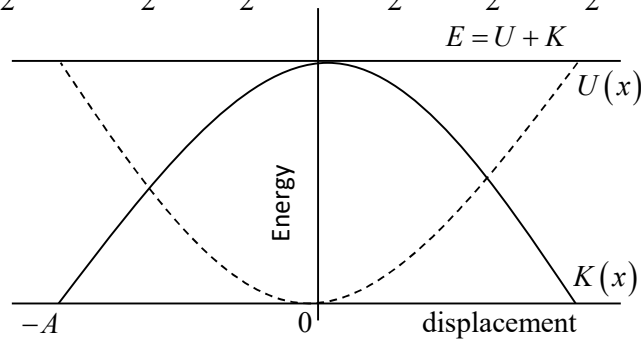
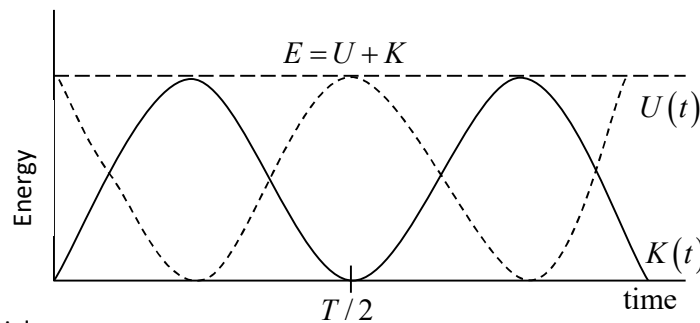


Figure: - Energy Diagram



Time average of potential energy

$$\langle V \rangle = \frac{\int_0^T V(t) dt}{\int_0^T dt} = \frac{\int_0^{\frac{2\pi}{\omega}} \frac{1}{2} kx^2 dt}{\int_0^{\frac{2\pi}{\omega}} dt} = \frac{\frac{1}{2} kA^2 \int_0^{\frac{2\pi}{\omega}} \cos^2(\omega t + \delta) dt}{\int_0^{\frac{2\pi}{\omega}} dt} = \frac{1}{4} kA^2$$

Time average of kinetic energy

$$\langle T \rangle = \frac{\int_0^T T(t) dt}{\int_0^T dt} = \frac{\int_0^{\frac{2\pi}{\omega}} \frac{1}{2} m\dot{x}^2 dt}{\int_0^{\frac{2\pi}{\omega}} dt} = \frac{\frac{1}{2} m\omega^2 A^2 \int_0^{\frac{2\pi}{\omega}} \sin^2(\omega t + \delta) dt}{\int_0^{\frac{2\pi}{\omega}} dt} = \frac{1}{4} m\omega^2 A^2 = \frac{1}{2} kA^2$$

Time average of total energy

$$\langle E \rangle = \langle T \rangle + \langle V \rangle = \frac{1}{4} kA^2 + \frac{1}{4} kA^2 = \frac{1}{2} kA^2$$

The expression for power of damped harmonic oscillator can be written as follows

$$\langle E \rangle = \langle T \rangle + \langle V \rangle = \frac{1}{4} kA^2 + \frac{1}{4} kA^2 = \frac{1}{2} kA^2$$

(a) Simple Pendulum

According to Hooke's law $F = -kx \Rightarrow ma = -kx \Rightarrow a = -\frac{k}{m}x = -\omega^2 x \Rightarrow \omega = \sqrt{\frac{k}{m}}$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

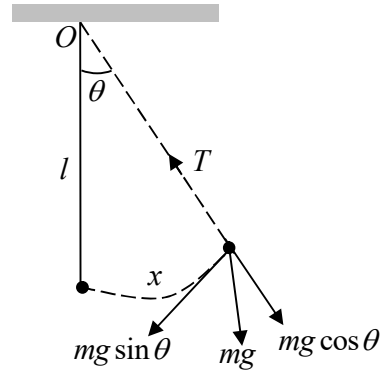
At equilibrium, $T = mg$

After displacement of θ , $T \cos(\theta) = mg$

Restoring force $F_R = -mg \sin(\theta)$

If θ is very small, then $\sin(\theta) = \theta = \frac{x}{l}$

$$F_R = -mg \frac{x}{l} = ma \Rightarrow a = g \frac{x}{l} = -\omega^2 x \Rightarrow \omega = \sqrt{\frac{g}{l}} \Rightarrow T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g}}$$



(b) Angular Simple Harmonic Motion

Angular Oscillation is the oscillation when a body is allowed to rotate freely about a given axis.

We know that $x = A \sin(\omega t + \phi)$

For angular simple harmonic motion

$$\Rightarrow l\theta = l\theta_0 \sin(\omega t + \phi), \text{ Where } -\theta_0 \leq \theta \leq \theta_0$$

The angular speed can be written as

$$\dot{\theta} = \frac{d\theta}{dt} = \omega \theta_0 \cos(\omega t + \phi),$$

$$\frac{d^2\theta}{dt^2} = -\omega^2 \theta_0 \sin(\omega t + \phi) = -\omega^2 \theta \Rightarrow \text{equation of motion}$$

$$\text{The kinetic energy } KE = \frac{I}{2} \dot{\theta}^2 = \frac{I}{2} \omega^2 \theta_0^2 \cos^2(\omega t + \phi) = \frac{I}{2} \omega^2 (\theta_0^2 - \theta^2)$$

$$\text{The maximum kinetic energy will be when } \theta = 0, KE_{\max} = \frac{I}{2} \omega^2 \theta_0^2 = \text{Total energy } (E_T)$$

$$\text{The potential energy } PE = E_T - KE = \frac{I}{2} \omega^2 \theta^2, PE = \frac{I}{2} \omega^2 \theta_0^2 \sin^2(\omega t + \phi)$$

$$\langle KE \rangle = \frac{I}{4} \omega^2 \theta_0^2 = \langle PE \rangle$$

(c) Compound Pendulum

Any rigid body suspended from a fixed support constitutes a physical pendulum. A circular ring suspended thorough on a nail in a wall, a heavy metallic rod suspended through hole in it are examples of compound pendulum.

The body rotates about a horizontal axis through O and perpendicular to the plane of motion. Let this axis be OA. Suppose the angular displacement of the body is θ at time t .

If l is the separation between point of suspension and the centre of mass, then torque can be written as

$$\tau = -mgl \sin(\theta)$$

Now if I is the moment of inertia about OA, and α is the angular acceleration, then

$$\tau = I\alpha = -mgl\theta \Rightarrow \alpha = \frac{-mgl\theta}{I} = -\omega^2\theta \Rightarrow \omega^2 = \frac{mgl}{I}$$

Time period, $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{mgl}}$

(d) Block in fluid

A Cylindrical wooden block of density σ cross section area A and length L floating up right when immersed to a depth d in liquid of density ρ as shown in figure. In the equilibrium Position the weight of block is balanced by upthrust force due to buoyancy. Block is slightly pressed vertically and released. show that motion of block is oscillatory.

Solution: The restoring force must be equal to weight of displaced liquid

$$F = -\rho Agy \Rightarrow m \frac{d^2y}{dt^2} = -\rho Agy$$

Where m is mass of block $m = \sigma Adg$ so equation of motion is given by $\sigma ALg \frac{d^2y}{dt^2} = -\rho Agy$

$$\frac{d^2y}{dt^2} + \frac{\rho}{\sigma L}y = 0 \text{ compare the equation with } \frac{d^2x}{dt^2} + \omega^2x = 0 \text{ the value of } \omega = \sqrt{\frac{\rho}{\sigma L}}.$$

(e) Capillary

Consider a U-shaped tube filled with liquid of density ρ up to height as shown in figure. initially the liquid is in equilibrium. The height of liquid in both arm of tube is L . If height of liquid is changing in one arm of tube with small distance y , then show that motion of oscillatory in nature and find angular frequency of oscillation.

If hight of liquid is increasing with height y in right arm of tube then it is also decreasing the y length in left arm. So, restoring force acting on liquid is $F = -\rho gA(2y)$ where A is area of cross section.

The equation of motion is given by $m \frac{d^2 y}{dt^2} = -\rho g A (2y)$.

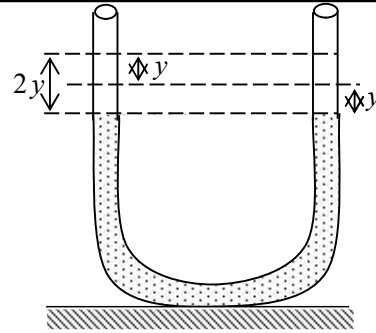
where m is mass of liquid

$m = \rho A g (2L)$ so, equation of motion will reduce to

$$\rho g A (2l) \frac{d^2 y}{dt^2} + \rho g A (2y) = 0$$

$$\rho g A (2l) \frac{d^2 y}{dt^2} + \rho g A (2y) = 0 \Rightarrow \frac{d^2 y}{dt^2} + \frac{g}{L} y = 0 \text{ comparing this equation with } \frac{d^2 x}{dt^2} + \omega^2 x = 0$$

So $\omega = \sqrt{\frac{g}{L}}$ and motion is considered as oscillatory.

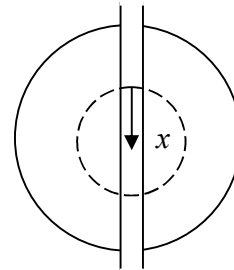


(f) Ball in tunnel of earth

There is a tunnel along the Diameter of earth. A ball of mass m is placed in the tunnel. Prove that motion of ball is equivalent to Simple harmonic motion find the angular frequency ball. Assume mass of earth is M and radius of earth is R .

Let us consider the mass density of earth is ρ if M is mass of earth then mass of inner part of sphere is

$$M_1 = \frac{M \rho \frac{4\pi}{3} x^3}{\frac{4\pi}{3} \rho R^3} = \frac{M x^3}{R^3}$$



$$F = -\frac{GM_1 m}{x^2} = -\frac{GM x^3}{R^3 \cdot x^2} = -\frac{GM x}{R^3} \Rightarrow m \frac{d^2 x}{dt^2} = -\frac{GM}{R^3} \cdot x \Rightarrow \frac{d^2 x}{dt^2} = -\frac{GM}{m R^3} x \text{ so } \omega = \sqrt{\frac{GM}{m R^3}}$$

(g) LC Circuit:

A capacitor of capacitance C is fully charged .it discharges through as inductor of inductance L connected through parallel with capacitor.

Prove that the discharging of charge in capacitor follow differential equation of small oscillation. Find the frequency of oscillation.

Solution let us assume current i is flowing in the circuit so the emf equation is given by

$$L \frac{di}{dt} + \frac{q}{c} = 0 \Rightarrow L \frac{d^2 q}{dt^2} + \frac{q}{c} = 0 \Rightarrow \frac{d^2 q}{dt^2} + \frac{q}{LC} = 0$$

If we compare above equation with $\frac{d^2 x}{dt^2} + \omega^2 x = 0$, $\omega = \sqrt{\frac{1}{LC}}$.

Damped Oscillator

A particle of mass m execute simple Harmonic oscillation with a constant amplitude. if the resultant force on it is proportional to the displacement and directed opposite to it. But nature provides a large number of situations in which damping or resistance force will act on it then the amplitude and frequency will dependent on damping parameter.

The damping force are in general complicated in nature but we can completely solve the problem if damping force is linear in velocity

Let us assume damping force is $-bv$ where b is damping coefficient.

The equation of motion is given by

$$m \frac{d^2 x}{dt^2} = -kx - bv \Rightarrow m \frac{d^2 x}{dt^2} = -kx - b \frac{dx}{dt} \Rightarrow \frac{d^2 x}{dt^2} + \frac{k}{m} x + \frac{b}{m} \frac{dx}{dt}$$

Where $\omega_0^2 = \frac{k}{m}$ and $2\beta = \frac{b}{m}$ where ω_0 is natural frequency of oscillator without damping. Also, β is the damping parameter.

So, equation of motion can be represented as second order differential equation

$$\frac{d^2 x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = 0 \dots\dots(1)$$

The general solution of equation of (1) is $x(t) = Ae^{\alpha t}$

$$Ae^{\alpha t} (\alpha^2 + 2\beta\alpha + \omega_0^2) = 0$$

$$\alpha = \frac{-2\beta \pm \sqrt{4\beta^2 - 4\omega_0^2}}{2}$$

$$\alpha_1 = -\beta + \sqrt{\beta^2 - \omega_0^2}, \alpha_2 = -\beta - \sqrt{\beta^2 - \omega_0^2}$$

$$x(t) = e^{-\beta t} \left(A_1 e^{(\sqrt{\beta^2 - \omega_0^2})t} + A_2 e^{(-\sqrt{\beta^2 - \omega_0^2})t} \right) \dots\dots\dots(2)$$

Cases:

- (1) $\beta = 0 \Rightarrow b = 0$, No drag force, (SHM)
- (2) $\beta^2 < \omega_0^2 \Rightarrow b^2 < 4mk$, oscillation (Under damping)
- (3) $\beta^2 > \omega_0^2 \Rightarrow b^2 > 4mk$, No oscillation (Over damping)
- (4) $\beta^2 = \omega_0^2 \Rightarrow b^2 = 4mk$, No oscillation (Critical damping)

Case-1

$\beta = 0 \Rightarrow b = 0$, No drag force, Ideal SHM

Now substitute $\beta = 0$ in equation (2), we will get

$$x(t) = C \cos \omega_0 t + D \sin \omega_0 t, \text{ where } C = A_1 + A_2, D = i(A_1 - A_2) \dots (3)$$

The equation (3) shows the SHM case where there is no damping.

Case -2

$\beta^2 < \omega_0^2 \Rightarrow b^2 < 4mk$, oscillation (Under damping)

Now let us define

$$\omega_1^2 = \omega_0^2 - \beta^2, A_1 = \frac{A}{2} e^{i\theta} \& A_2 = \frac{A}{2} e^{-i\theta}$$

Now, substituting these in equation (2), we will get

$$x(t) = Ae^{-\beta t} \cos(\omega_1 t + \theta) \dots (4)$$

$$A(t) = Ae^{-\beta t} = \text{Amplitude} \Rightarrow \text{Decaying function}, \cos(\omega_1 t + \theta) = \text{Oscillating function}$$

It is clearly noticeable from equation (4) that there is oscillation but the amplitude is exponentially decaying with time. The amplitude is decaying due to loss of energy caused by energy dissipation.

In under damping case,

$$\text{We know that } A(t) = Ae^{-\beta t}, \text{ Also, } E(t) = A(t)^2 = A^2 e^{-2\beta t} = E_0 e^{-2\beta t}$$

Thus, energy is also decaying with time.

Relaxation time (τ) : The time in which energy stored in the system reduces to $1/e$ of its initial value is known as relaxation time.

$$E(t) = E_0 e^{-2\beta t} = \frac{E_0}{e}$$

$$e^{-2\beta t} = e^{-1} \Rightarrow \tau = \frac{1}{2\beta}$$

Rate of energy dissipation (P)/Power:

$$P = -\frac{dE}{dt} = 2\beta E(t) = \frac{b}{m} E(t) = \frac{E(t)}{\tau}$$

Quality factor (Q): $Q = 2\pi \times \frac{\text{Energy stored}}{\text{Energy loss per cycle}} = 2\pi \times \frac{E}{\frac{E}{T}} = \omega_0 \tau = \omega_0 \frac{m}{b}$

Case -3 Over damping

$$\beta^2 > \omega_0^2 \Rightarrow b^2 > 4mk, \text{ No oscillation (Over damping)}$$

Now let us define $\omega_2^2 = \beta^2 - \omega_0^2$

Now, substitute this in equation (2), we will get

$$x(t) = e^{-\beta t} (A_1 e^{\omega_2 t} + A_2 e^{-\omega_2 t}) = (A_1 e^{-(\beta - \omega_2)t} + A_2 e^{-(\beta + \omega_2)t}) \dots (5)$$

We can see that there is no oscillation. Also, the amplitude is decaying exponentially.

Case -4 Critical damping

$$\beta^2 = \omega_0^2 \Rightarrow b^2 = 4mk, \text{ Critical damping}$$

Describe the trajectory under critical damping, let us start with original differential equation

$$\frac{d^2 x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = 0$$

$$\frac{d^2 x}{dt^2} + 2\beta \frac{dx}{dt} + \beta^2 x = 0 \quad [\text{since, } \beta = \omega_0]$$

$$\left(\frac{d}{dt} + \beta\right)\left(\frac{d}{dt} + \beta\right)x = 0, \quad \text{let } \left(\frac{d}{dt} + \beta\right)x = y$$

$$\left(\frac{d}{dt} + \beta\right)y = 0$$

$$\int \frac{dy}{y} = \int -\beta dt \Rightarrow \ln y = -\beta t \Rightarrow y = A e^{-\beta t} \quad [A = \text{constant}]$$

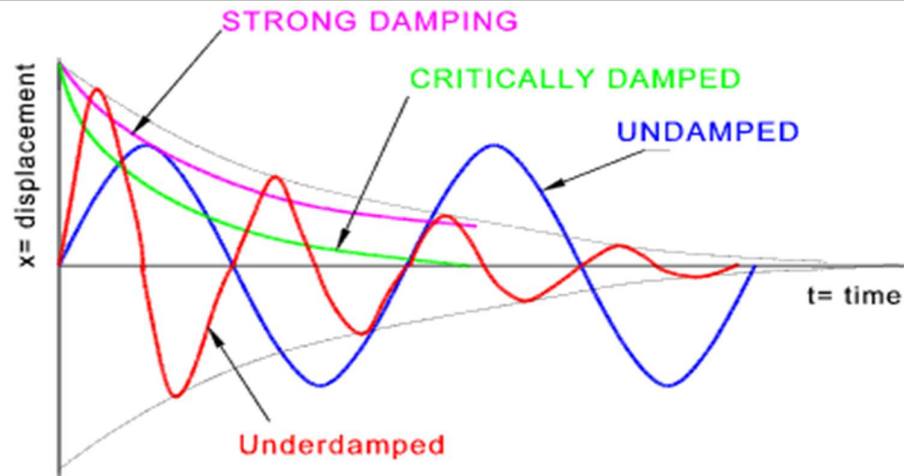
$$\left(\frac{d}{dt} + \beta\right)x = A e^{-\beta t}$$

$$\left(\frac{d}{dt} + \beta\right)x e^{\beta t} = A$$

$$\left(\frac{d}{dt} x e^{\beta t}\right) = A$$

$$x = (At + B)e^{-\beta t}$$

Since, the exponential term is dominating. Thus, there is a rapid decaying in amplitude. Also, there is no oscillation under critical damping.



Forced Oscillation

In this case, the oscillation is driven by some oscillating force

Let us consider $F_0 \sin(pt)$ is the external force which is applied to drive the oscillation. Here,

p is representing the driving frequency

Now the equation of motion under the presence of restoring force, damping force and external force is

$$F_{net} = -kx - b \frac{dx}{dt} + F_0 \sin(pt)$$

$$-b \frac{dx}{dt} = \text{Damping force}$$

Where, $-kx$ = restoring force

$F_0 \sin(pt)$ = External periodic force

$$ma = -kx - b \frac{dx}{dt} - F_0 \sin(pt)$$

$$\frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = \frac{F_0}{m} \sin(pt)$$

$$\frac{d^2x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = f_0 \sin(pt) \dots (1) \quad 2\beta = b/m, \omega_0^2 = k/m, f_0 = F_0/m$$

The general solution of equation (1) is $x = A \sin(pt - \theta)$, Substitute this in equation (1), we will get

$$-p^2 A \sin(pt - \theta) + 2\beta p A \cos(pt - \theta) + \omega_0^2 A \sin(pt - \theta) = f_0 \sin(pt - \theta + \theta)$$

$$-p^2 A \sin(pt - \theta) + 2\beta p A \cos(pt - \theta) + \omega_0^2 A \sin(pt - \theta)$$

$$= f_0 (\sin(pt - \theta) \cos \theta + \cos(pt - \theta) \sin \theta) \dots (2)$$

Now, comparing the coefficient of $\sin \theta$ and $\cos \theta$, we will get

$$(\omega_0^2 - p^2) A = f_0 \cos(\theta) \dots (3)$$

$$2p\beta A = f_0 \sin(\theta) \dots (4)$$

$$\tan(\theta) = \frac{2p\beta}{(\omega_0^2 - p^2)}$$

From equation (3) and (4), we can write

$$A = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2}} \dots\dots(5)$$

$$x(t) = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2}} \sin(pt - \theta) \dots\dots(6)$$

Case 1

At very low driving frequency, $p \ll \omega_0$

We will get the amplitude from equation (5)

$$A = \frac{f_0}{\omega^2} = \frac{F_0}{K}$$

Thus, A depends on force constant A but independent on driving frequency.

$$x(t) = \frac{F_0}{K} \sin(pt - \theta)$$

Case 2

At very high driving frequency, $p \gg \omega_0$

$$A = \frac{f_0}{p^2} = \frac{F_0}{mp^2}$$

Thus, A depends on driving frequency.

Amplitude Resonance:

The value of driving frequency p at which amplitude become maximum is called amplitude resonance.

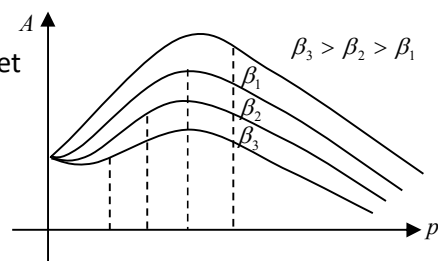
From, equation (5), it is clearly noticeable that $A = A_{\max}$, when

$$\frac{d}{dp} \left[(\omega_0^2 - p^2)^2 + 4\beta^2 p^2 \right] = 0 \Rightarrow -4\omega_0^2 + 4p^2 + 8\beta^2 = 0 \Rightarrow p = \sqrt{\omega_0^2 - 2\beta^2}$$

Now, if we substitute $p = \sqrt{\omega_0^2 - 2\beta^2}$ in equation (5), we will get

$$A_{\max} = \frac{f_0}{\sqrt{4\beta^4 + 4\beta^2(\omega_0^2 - 2\beta^2)}}$$

$$A_{\max} = \frac{f_0}{\sqrt{4\beta^2\omega_0^2 - 4\beta^4}} = \frac{f_0}{2\beta\sqrt{\omega_0^2 - \beta^2}}$$



Velocity

$$x(t) = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2}} \sin(pt - \theta)$$

The expression for velocity can be written as,

$$v(t) = \frac{dx(t)}{dt} = \frac{f_0 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2}} \cos(pt - \theta) = v_A \cos(pt - \theta)$$

$$v_A = \frac{f_0 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2}}$$

Case 1

At very low driving frequency, $p \ll \omega_0$

$$v_A = \frac{f_0 p}{\omega_0^2} = \frac{F_0 p}{K}$$

Case 2

At very high driving frequency, $p \gg \omega_0$

$$v_A = \frac{f_0 p}{p^2} = \frac{F_0}{mp}$$

Expression for maximum velocity

$$v_A = \frac{f_0}{\sqrt{\frac{(\omega_0^2 - p^2)^2}{p^2} + 4\beta^2}}$$

$$v_A = v_A^{\max}, \text{ when } p = \omega_0, v_A^{\max} = \frac{f_0}{2\beta}$$

Power dissipated by oscillator

$$P = -F_d v = b \left(\frac{dx}{dt} \right)^2 = \frac{b f_0^2 p^2}{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2} \cos^2(pt - \theta)$$

$$\text{The average power will be, } P_{av} = \frac{b f_0^2 p^2}{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2} \langle \cos^2(pt - \theta) \rangle$$

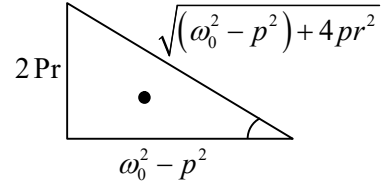
$$= \frac{1}{2} \frac{b f_0^2 p^2}{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2} = \frac{m \beta f_0^2}{\frac{(\omega_0^2 - p^2)^2}{p^2} + 4\beta^2}$$

$$P_{av}^{\max} = \frac{mf_0^2}{4\beta} \quad (\omega = p_0)$$

Power absorbed by oscillator

$$P = F_0 \sin(pt) v = \frac{F_0 f_0 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2}} \sin(pt) \cos(pt - \theta)$$

$$= \frac{F_0 f_0 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2}} \left[\frac{\sin(2pt)}{2} \cos(\theta) + \sin^2(pt) \sin(\theta) \right]$$



The average power will be

$$P_{av} = \frac{F_0 f_0 \beta p^2}{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2} = \frac{F_0 f_0 \beta}{\frac{(\omega_0^2 - p^2)^2}{p^2} + 4\beta^2}$$

The maximum average power will be, $P_{av}^{\max} = \frac{mf_0^2}{4\beta} \quad (\omega = p_0)$

Band width of resonance

The difference in values of driving frequency at which the power absorbed drops to half of its maximum value is known as band width resonance.

$$\frac{F_0 f_0 \beta p^2}{(\omega_0^2 - p^2)^2 + 4\beta^2 p^2} = \frac{mf_0^2}{8\beta} \Rightarrow (\omega_0^2 - p^2)^2 + 4\beta^2 p^2 = 8\beta^2 p^2$$

$$\Rightarrow (\omega_0^2 - p^2) = \pm 2\beta p$$

$$\Rightarrow (\omega_0 - p) = \pm \frac{2\beta p}{(\omega_0 + p)} = \pm \frac{2\beta}{\left(\frac{\omega_0}{p} + 1\right)} = \pm \frac{2\beta}{(1+1)} = \pm \beta \quad (\omega_0 \approx p)$$

$$\Rightarrow (\omega_0 - \omega_1) = \beta$$

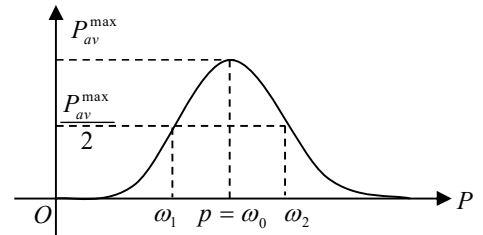
$$\Rightarrow (\omega_0 - \omega_2) = -\beta$$

$$\Rightarrow (\omega_0 - \omega_1) = \beta$$

$$\Rightarrow (\omega_2 - \omega_1) = 2\beta = \frac{1}{\tau}$$

Quality factor (Q):

$$Q = \frac{1}{2} \left[1 + \left(\frac{\omega_0}{p} \right)^2 \right] (p\tau)$$



Superposition of SHMs

Case-1 Addition of two SHMs having same frequency but different amplitude.

In this case we can write the SHMs equations as

$$y_1(t) = a_1 \sin(\omega t + \delta_1) \text{ \& } y_2(t) = a_2 \sin(\omega t + \delta_2)$$

The resultant amplitude can be written as

$$y(t) = y_1(t) + y_2(t) = a_1 \sin(\omega t + \delta_1) + a_2 \sin(\omega t + \delta_2)$$

$$y(t) = \sin(\omega t) \{a_1 \cos(\delta_1) + a_2 \cos(\delta_2)\} + \cos(\omega t) \{a_1 \sin(\delta_1) + a_2 \sin(\delta_2)\} \dots (1)$$

Let us assume,

$$\{a_1 \cos(\delta_1) + a_2 \cos(\delta_2)\} + \cos(\omega t) \{a_1 \sin(\delta_1) + a_2 \sin(\delta_2)\}$$

$$a_1 \cos(\delta_1) + a_2 \cos(\delta_2) = A \cos(\theta) \dots (2)$$

$$a_1 \sin(\delta_1) + a_2 \sin(\delta_2) = A \sin(\theta) \dots (3)$$

Now, substitute equation (2) and (3) in (1), we will get

$$y(t) = A \sin(\omega t + \theta) \dots (4)$$

It is clearly noticeable that the resultant wave is SHM.

The amplitude A can be calculated from equations (2) and (3)

$$A = \sqrt{(a_1^2 + a_2^2 + 2a_1a_2 \cos(\delta_1 - \delta_2))}$$

$$A_{\max} = a_1 + a_2, \text{ when } \cos(\delta_1 - \delta_2) = 2n\pi, n = 0, 1, 2, 3, \dots$$

$$A_{\min} = a_1 - a_2, \text{ when } \cos(\delta_1 - \delta_2) = (2n+1)\pi, n = 0, 1, 2, 3, \dots$$

$$\tan(\theta) = \frac{a_1 \sin(\delta_1) + a_2 \sin(\delta_2)}{a_1 \cos(\delta_1) + a_2 \cos(\delta_2)}$$

Case-2 Addition of two SHMs having same amplitude but different frequency

$$y_1(t) = a \sin(\omega_1 t) \text{ \& } y_2(t) = a \sin(\omega_2 t)$$

The resultant wave can be written as

$$y(t) = y_1(t) + y_2(t) = a \sin(\omega_1 t) + a \sin(\omega_2 t) = 2a \cos \frac{(\omega_1 - \omega_2)}{2} t \sin \frac{(\omega_1 + \omega_2)}{2} t$$

$$y(t) = y_1(t) + y_2(t) = a \sin(\omega_1 t) + a \sin(\omega_2 t) = 2a \cos \frac{(\omega_1 - \omega_2)}{2} t \sin \frac{(\omega_1 + \omega_2)}{2} t$$

$$y(t) = A(t) \sin \frac{(\omega_1 + \omega_2)}{2} t \Rightarrow \text{Resultant wave is periodic but not SHM}$$

Where $A(t) = 2a \cos \frac{(\omega_1 - \omega_2)}{2} t$ = Amplitude

We know that the intensity, $I \propto \text{Amplitude}^2$

Thus, $I = 4a^2 \cos^2 \frac{(\omega_1 - \omega_2)}{2} t$

Intensity maximum

$$\cos^2 \frac{(\omega_1 - \omega_2)}{2} t = 1 \Rightarrow \cos \frac{(\omega_1 - \omega_2)}{2} t = \cos n\pi$$

$$\frac{(\omega_1 - \omega_2)}{2} t = n\pi, n = 0, 1, 2, 3$$

$$\frac{(2\pi n_1 - 2\pi n_2)}{2} t = n\pi, n = 0, 1, 2, 3$$

$$t = 0, \frac{1}{n_1 - n_2}, \frac{2}{n_1 - n_2}, \dots$$

Time difference between to maximum intensity $\Delta t = \frac{1}{n_1 - n_2}$

Thus, the beat frequency $f_b = n_1 - n_2$

Intensity minima

$$\cos^2 \frac{(\omega_1 - \omega_2)}{2} t = 0 \Rightarrow \cos \frac{(\omega_1 - \omega_2)}{2} t = \cos (2n + 1) \frac{\pi}{2}$$

$$\frac{(2\pi n_1 - 2\pi n_2)}{2} t = (2n + 1) \frac{\pi}{2}, n = 0, 1, 2, 3$$

$$t = \frac{1}{2} \frac{1}{n_1 - n_2}, \frac{3}{2} \frac{1}{n_1 - n_2}, \dots$$

Time difference between to minimum intensity $\Delta t = \frac{1}{n_1 - n_2}$

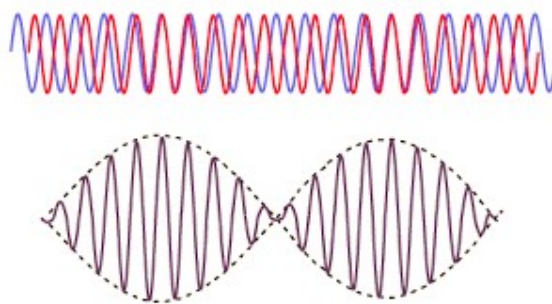
Thus, the beat frequency $f_b = n_1 - n_2$

Case-3 Addition of two SHMs having different amplitude but different frequency

$$y_1(t) = a_1 \sin(\omega_1 t) \text{ \& } y_2(t) = a_2 \sin(\omega_2 t)$$

$$y_1(t) = a_1 \sin\left(\frac{\omega_1 + \omega_2}{2} t + \frac{\omega_1 - \omega_2}{2} t\right), y_2(t) = a_2 \sin\left(\frac{\omega_1 + \omega_2}{2} t - \frac{\omega_1 - \omega_2}{2} t\right),$$

Let, $\frac{\omega_1 + \omega_2}{2} = \omega_{av} \Rightarrow \frac{\omega_1 - \omega_2}{2} = \omega_m$



$$y_1(t) = a_1 \sin(\omega_{av} + \omega_m)t, \quad y_2(t) = a_2 \sin(\omega_{av} - \omega_m)t,$$

The resultant wave can be written as follows

$$\begin{aligned} y(t) &= y_1(t) + y_2(t) = a_1 \sin(\omega_{av} + \omega_m)t + a_2 \sin(\omega_{av} - \omega_m)t \\ &= \sin \omega_{av}t (a_1 \cos \omega_m t + a_2 \cos \omega_m t) + \cos \omega_{av}t (a_1 \sin \omega_m t + a_2 \sin \omega_m t) = A \sin(\omega_{av}t + \theta) \end{aligned}$$

$$\text{Where, } A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos(2\omega_m t)}, \quad \tan \theta = \left(\frac{a_1 - a_2}{a_1 + a_2} \right) \tan(\omega_m t)$$

Case-4 Addition of N number of SHMs having same amplitude but different frequency

$$y_1(t) = a \cos(\omega t), \quad y_2(t) = a \cos(\omega + \delta\omega)t, \quad y_3(t) = a \cos(\omega + 2\delta\omega)t, \dots, y_N(t) = a \cos(\omega_2)t$$

We can write

$$\delta\omega = \frac{\omega_2 - \omega_1}{N-1}$$

The resultant amplitude can be written as follows

$$\begin{aligned} y(t) &= y_1(t) + y_2(t) + y_3(t) + \dots + y_N(t) \\ &= a \cos(\omega t) + a \cos(\omega + \delta\omega)t + a \cos(\omega + 2\delta\omega)t + \dots + a \cos(\omega_2)t \\ &= a \operatorname{Re} \left[e^{i\omega_1 t} \{1 + e^{i\delta\omega t} + e^{i2\delta\omega t} + \dots + e^{i(N-1)\delta\omega t}\} \right] \\ &= a \operatorname{Re} \left[e^{i\omega_1 t} \frac{e^{\frac{iN\delta\omega t}{2}} - e^{\frac{i\delta\omega t}{2}}}{e^{\frac{i\delta\omega t}{2}} - e^{-\frac{i\delta\omega t}{2}}} \right] = a \frac{\sin(\frac{N\delta\omega t}{2})}{\sin(\frac{\delta\omega t}{2})} \cos \frac{(\omega_1 + \omega_2)t}{2} \end{aligned}$$

Case-5 Lissajous figures

(Particle is subjected to two mutually perpendicular SHMs simultaneously)

We can write the equation

$$x = a \sin(\omega t + \phi), \quad y = b \sin(\omega t)$$

$$x = a \sin(\omega t) \cos(\phi) + a \cos(\omega t) \sin(\phi) = a(\sin(\omega t) \cos(\phi) + \cos(\omega t) \sin(\phi))$$

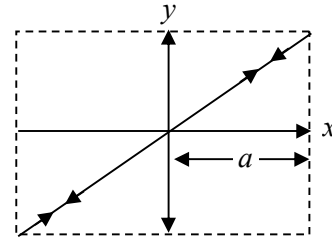
$$\frac{x}{a} = \left\{ \frac{y}{b} \cos(\phi) + \sqrt{1 - \left(\frac{y}{b}\right)^2} \sin(\phi) \right\}$$

$$\left[\frac{x}{a} - \frac{y}{b} \cos(\phi) \right]^2 = \left\{ \sqrt{1 - \left(\frac{y}{b}\right)^2} \sin(\phi) \right\}^2$$

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 - \frac{2xy}{ab} \cos(\phi) = \sin^2 \phi \dots \dots \dots (1)$$

Case-1 $\phi = 0^\circ$, $y = \frac{b}{a}x \Rightarrow$ Equation of straight line

The trajectory will be straight line



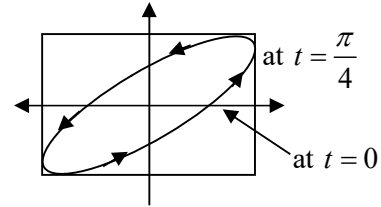
Case-2 $\phi = 45^\circ$, $\sin \phi = \cos \phi = 1/\sqrt{2}$

$$x = a \sin\left(\omega t + \frac{\pi}{4}\right)$$

$$y = b \sin(\omega t)$$

At $\omega t = 0$, $x = \frac{a}{\sqrt{2}}$, $y = 0$

At $\omega t = \frac{\pi}{4}$, $x = a$, $y = \frac{b}{\sqrt{2}}$



The trajectory of motion is oblique ellipsoid in figure below

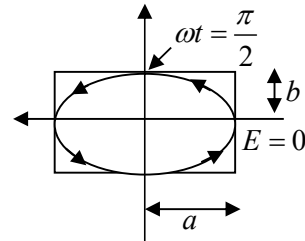
Case-3 $\phi = \frac{\pi}{2}$, $\sin \phi = 1$, $\cos \phi = 0$

$$x = a \sin\left(\omega t + \frac{\pi}{2}\right)$$

$$y = b \sin(\omega t)$$

At $\omega t = 0$, $x = a$, $y = 0$

At $\omega t = \frac{\pi}{4}$, $x = \frac{a}{\sqrt{2}}$, $y = \frac{b}{\sqrt{2}}$



The trajectory of motion is shown ellipsoid

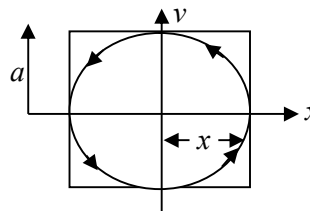
Case-4 When $\phi = \frac{\pi}{2}$, and $a = b$

$$\sin \phi = 1, \cos \phi = 0$$

$$x = a \sin\left(\omega t + \frac{\pi}{2}\right), y = a \sin(\omega t)$$

At $\omega t = 0$, $x = a$, $y = 0$

At $\omega t = \frac{\pi}{4}$, $x = \frac{a}{\sqrt{2}}$, $y = \frac{a}{\sqrt{2}}$



The trajectory of motion is shown below

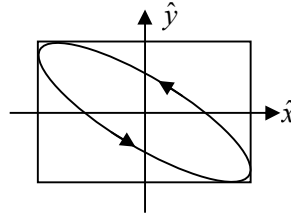
Case-5 When $\phi = \frac{3\pi}{4}$

$$x = a \sin\left(\omega t + \frac{3\pi}{4}\right), \quad y = b \sin(\omega t)$$

At $\omega t = 0$, $x = \frac{a}{\sqrt{2}}$, $y = 0$

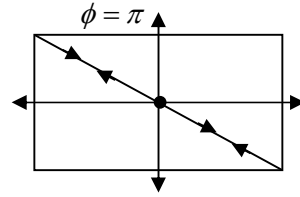
At $\omega t = \frac{\pi}{4}$, $x = 0$, $y = \frac{b}{\sqrt{2}}$

The trajectory of motion is shown below

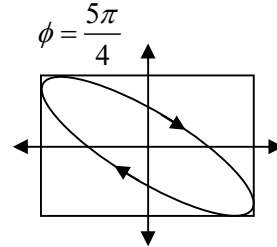


Case-6 $\phi = \pi$, $y = -\frac{b}{a}x \Rightarrow$ Equation of straight line

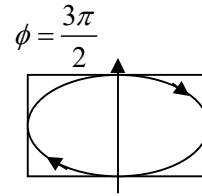
The trajectory is shown below.



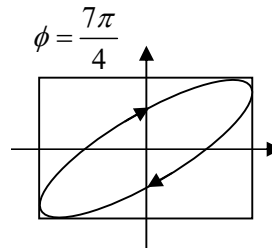
Case-7 $\phi = \frac{5\pi}{4}$



Case 8: $\phi = \frac{3\pi}{2}$ -Direction will be opposite



Case 9: $\phi = \frac{7\pi}{4}$ -Direction will be opposite



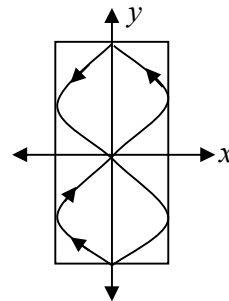
Now consider the following situation

$$x = a \sin(2\omega t + \phi)$$

$$y = b \sin(\omega t)$$

The general solution of these two equations will provides

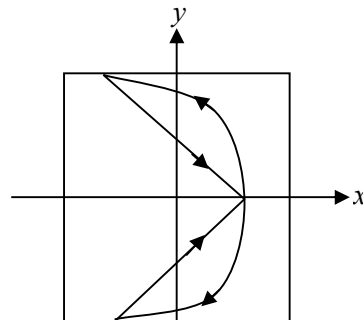
$$\left\{\frac{x}{a} - \sin(\phi)\right\}^2 + 4\frac{y^2}{b^2}\left(\frac{y^2}{b^2} - 1 + \frac{x}{a}\cos\phi\right)$$



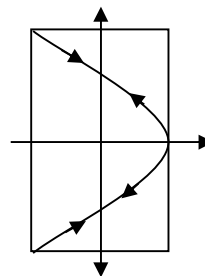
The trajectory of motion is show for different phase

When $\phi = 0$

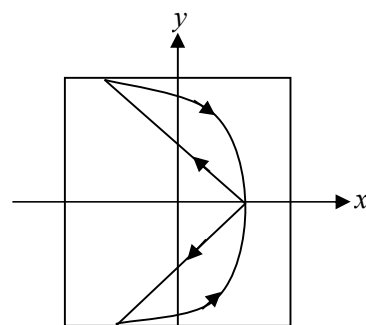
When $\phi = \frac{\pi}{4}$



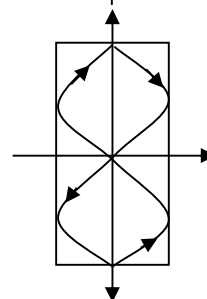
When $\phi = \frac{\pi}{2}$



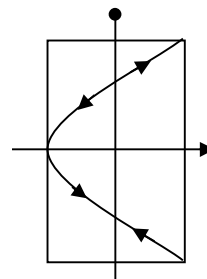
When $\phi = \frac{3\pi}{4}$



When $\phi = \pi$



When $\phi = \frac{3\pi}{2}$



Wave Motion

A wave is a disturbance in a medium that carries energy without a net movement of particles. It may take the form of elastic deformation, a variation of pressure, electric or magnetic intensity, electric potential, or temperature.

About wave motion

- Transfers energy.
- Usually involves a periodic, repetitive movement.
- Does not result in a net movement of the medium or particles in the medium (mechanical wave).

Few basic terminology

(a) Wavelength (λ)

Wavelength is the distance between two successive identical parts of the wave.

(a) Amplitude (A)

Amplitude is the maximum displacement from the neutral position. This represents the energy of the wave. Greater amplitude carries greater energy. Displacement is the position of a particular point in the medium as it moves as the wave passes. Maximum displacement is the amplitude of the wave

(c) Frequency (f)

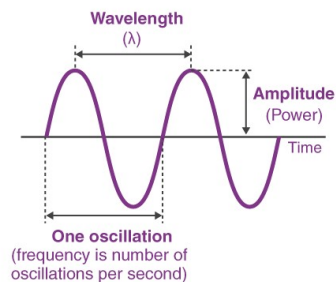
Frequency (f) is the number of repetitions per second in Hz.

(d) Time period (T)

Period (T) is the time for one wavelength to pass a point.

(e) Velocity (v)

The velocity (v) of the wave is the speed at which a specific part of the wave passes a point. The speed of a light wave is c .



Types of Waves:

The types of waves are given below.

Transverse Waves

Waves in which the medium moves at right angles to the direction of the wave.

Examples of transverse waves:

- Water waves (ripples of gravity waves, not sound through water)

- Light waves
- S-wave earthquake waves
- Stringed instruments
- Torsion wave

The high point of a transverse wave is a crest. The low part is a trough.

Longitudinal Wave:

A longitudinal wave has the movement of the particles in the medium in the same dimension as the direction of movement of the wave.

Examples of longitudinal waves:

- Sound waves
- P-type earthquake waves
- Compression wave

The displacement of medium particle can be written as follows

$$y = A \sin(\omega t - kx)$$

A = Amplitude, ω = Angular frequency,

$$k = \frac{2\pi}{\lambda} = \text{Wave vector, } \lambda = \text{Wavelength}$$

Or $y = \sin \frac{2\pi}{\lambda} (vt - x)$ where $v = \frac{\omega}{k} = \text{wave velocity}$

velocity = Phase velocity

$\Delta\phi$ = Phase difference

Δx = Path difference

Δt = Time difference

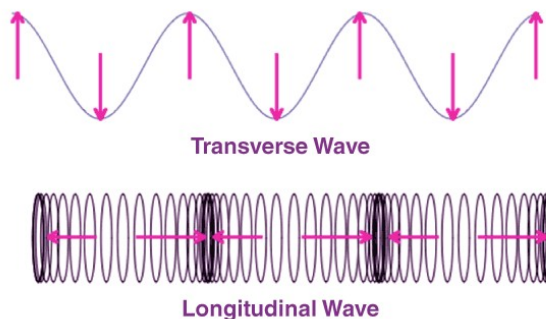
Important relation

$$\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda} = \frac{\Delta t}{T}$$

Thus, $\Delta\phi = \frac{2\pi}{\lambda} \Delta x$

Particle velocity (v_p): The rate of change of displacement with respect to equilibrium.

$$v_p = \frac{dy}{dt} = -v (\text{Phase velocity}) \times \text{Slope of } y \text{ vs } x \text{ curve}$$



$$\frac{dy}{dt} = -\frac{\omega}{k} \times \frac{dy}{dx}$$

We know that,

$$y = A \sin(\omega t - kx)$$

$$v_p = \frac{dy}{dt} = A\omega \cos(\omega t - kx)$$

Acceleration of medium particle (a_p): **Acceleration** (symbol: (a) is defined as the rate of change (or time derivative) of velocity.

$$a_p = \frac{d^2 y}{dt^2}$$

$$a_p = -A\omega^2 \sin(\omega t - kx)$$

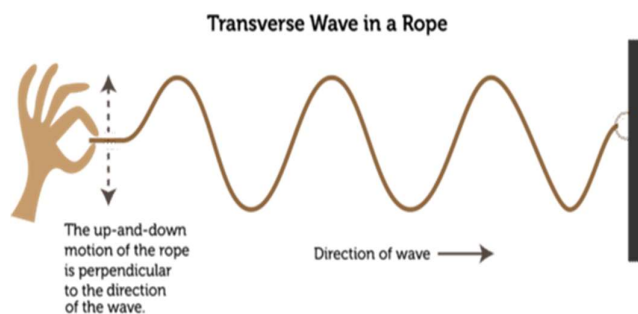
The velocity of sound waves can be written as

$$a_p = -A\omega^2 \sin(\omega t - kx)$$

Wave Motion

Speed of transverse wave in string:

$$v = \sqrt{\frac{T}{\mu}}; \quad T = \text{Tension}, \mu = \text{Mass per unit length}$$

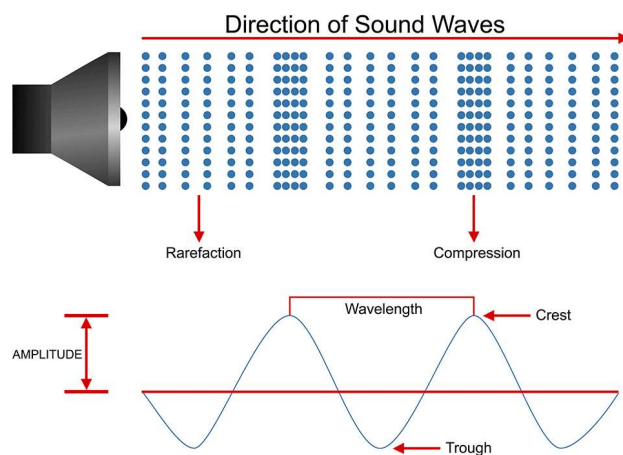


Speed of transverse wave in string:

$$y = A \sin(\omega t - kx)$$

The velocity of sound wave can be written as

$$v = \sqrt{\frac{E}{\rho}}, \quad E = \text{Modulus of elasticity}, \rho = \text{Density of medium}$$



For Solid

$E = Y = \text{Young modulus}, Y = \text{Stress/Strain}$

$$v = \sqrt{\frac{Y}{\rho}}$$

For liquid

$E = B = \text{Bulk modulus}, B = \text{Excess pressure/Volume strain}$

$$B = -v \frac{\Delta p}{\Delta v}$$

$$v = \sqrt{\frac{B}{\rho}}$$

Velocity of sound in glass/air

Newtons assume that velocity propagation is isothermal process.

$$pv = \text{constant}$$

$$vdp + p dv = 0$$

$$\Rightarrow p = -\frac{vdp}{dv} = B \Rightarrow v = \sqrt{\frac{p}{\rho}}$$

In air $p = 1.01 \times 10^5 \text{ Pa}$, $\rho = 1.293 \text{ kg/m}^3$

$$v = 279.43 \text{ m/s}$$

But in experiment, $v = 332 \text{ m/s}$

Later the correction is made by Laplace. He assumed that the speed of sound in air/glass is adiabatic process.

$$pv^\gamma = \text{constant}$$

$$v^\gamma dp + \gamma p v^{\gamma-1} dv = 0 \Rightarrow p = -\frac{vdp}{dv} = B$$

$$\text{Thus, } v_{\text{sound}} = \sqrt{\frac{\gamma p}{\rho}} = 332 \text{ m/s}$$

We know that

$$PV = nkT, n = \frac{m}{M} N_A$$

$$P = \frac{m}{VM} N_A kT \Rightarrow \frac{P}{\rho} = \frac{RT}{M} \Rightarrow v_{\text{sound}} = \sqrt{\frac{\gamma RT}{M}}$$

If medium is same but different temperature

$$\frac{v_{\text{sound1}}}{v_{\text{sound2}}} = \sqrt{\frac{T_1}{T_2}}$$

If medium is different

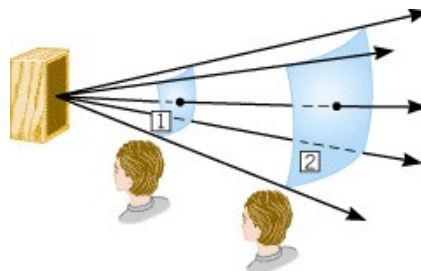
$$\frac{v_{\text{sound}_1}}{v_{\text{sound}_2}} = \sqrt{\frac{\gamma_1 T_1 M_2}{\gamma_2 T_2 M_1}}$$

Power of wave (P)

The power is energy transported through a perpendicular surface of area a in unit time.

The expression for power can be written as follows

$$P = 2\pi^2 n^2 A^2 \rho v s$$



n = Frequency of wave

A = Amplitude of wave

ρ = Density of medium

v = Velocity

s = Surface area

Again, energy stored per unit volume = $2\pi^2 n^2 A^2 \rho$

Power transported by string wave

$$P = 2\pi^2 n^2 A^2 \rho v s$$

$$\rho s = \frac{m}{l} s = m/l = \mu \text{ Linear mass density}$$

$$P = 2\pi^2 n^2 A^2 \mu v$$

Energy transported along string in one period

$$E = PT = 2\pi^2 n^2 A^2 \mu v \frac{1}{n} \Rightarrow E = PT = 2\pi^2 n A^2 \mu v$$

Wave intensity

Power transported per unit correctional area is called intensity

$$I = \frac{P}{s} = 2\pi^2 n^2 A^2 \rho v$$

Representing sound wave as a pressure wave

We know that the displacement of wave can be written as follows

$$y = A \sin(\omega t - kx)$$

$$dy = -kA \cos(\omega t - kx) dx$$

Volume compression $\Delta V = -kA \cos(\omega t - kx) dx$

$$\frac{\Delta V}{V} = -kA \cos(\omega t - kx)$$

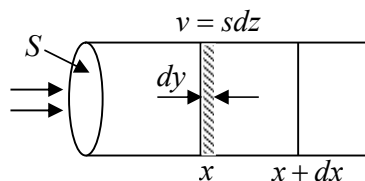
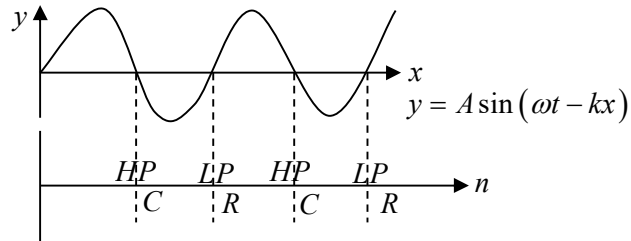
$$\text{Now, } B = -V \frac{\Delta P}{\Delta V} = BkA \cos(\omega t - kx)$$

$$\Delta p = BkA \cos(\omega t - kx)$$

$$\Delta p = \Delta p_0 \cos(\omega t - kx);$$

Δp_0 = Pressure amplitude

$$\Delta p_0 = BKA = BA \frac{2\pi}{\lambda} \Rightarrow A = \frac{\lambda \Delta p_0}{2\pi B}$$



Intensity of sound wave

$$I = 2\pi^2 n^2 A^2 \rho v; \quad v = n\lambda; \quad v = \sqrt{\frac{B}{\rho}} \Rightarrow B = \rho v^2$$

$$I = 2\pi^2 \left(\frac{v}{\lambda}\right)^2 \left(\frac{\lambda \Delta p_0}{2\pi B}\right)^2 \rho v; \quad B = \rho v^2$$

$$I = \frac{(\Delta p_0)^2}{2\rho v}; \quad v = \text{Wave velocity}$$

Wave Motion

Pitch: It is the perception of sound frequency by human ears.

Higher Frequency Sound: High pitch

Low Frequency Sound: High pitch



(a) Audible frequency: $f: 20 \text{ Hz} - 20 \text{ KHz}$

(b) Infrasonic frequency: $f < 20 \text{ Hz}$

(c) Ultrasonic frequency $f > 20 \text{ KHz}$

Loudness of sound waves (L)

It is the perception of human mind towards its intensity.

Let us assume I_0 is the lower limit of human hearing.

$$I_0 = 10^{-12} \text{ W / m}^2$$

The expression of loudness (L) can be written as

$$L(\text{dB}) = 10 \log_{10} \left(\frac{I_1}{I_0} \right)$$

The relative loudness can be written as

$$L(\text{dB}) = 10 \log_{10} \left(\frac{I_1}{I_2} \right)$$

Standing waves:

$$y_1 = A \sin(\omega t - kx), \quad y_2 = -A \sin(\omega t + kx)$$

The resultant wave can be written as

$$y = y_1 + y_2 = A \sin(\omega t - kx) - A \sin(\omega t + kx)$$

$$y = y_1 + y_2 = -2A \sin(kx) \cos(\omega t)$$

$$y = -2A \sin\left(\frac{2\pi}{\lambda} x\right) \cos\left(\frac{2\pi}{\lambda} vt\right)$$

The above equation represents the standing wave. Net energy transfer is zero.

$$y_1 = A \sin(\omega t - kx), \quad y_2 = A \sin(\omega t + kx)$$

$$y = y_1 + y_2 = 2A \cos\left(\frac{2\pi}{\lambda} x\right) \sin\left(\frac{2\pi}{\lambda} vt\right)$$

Energy of stationary wave

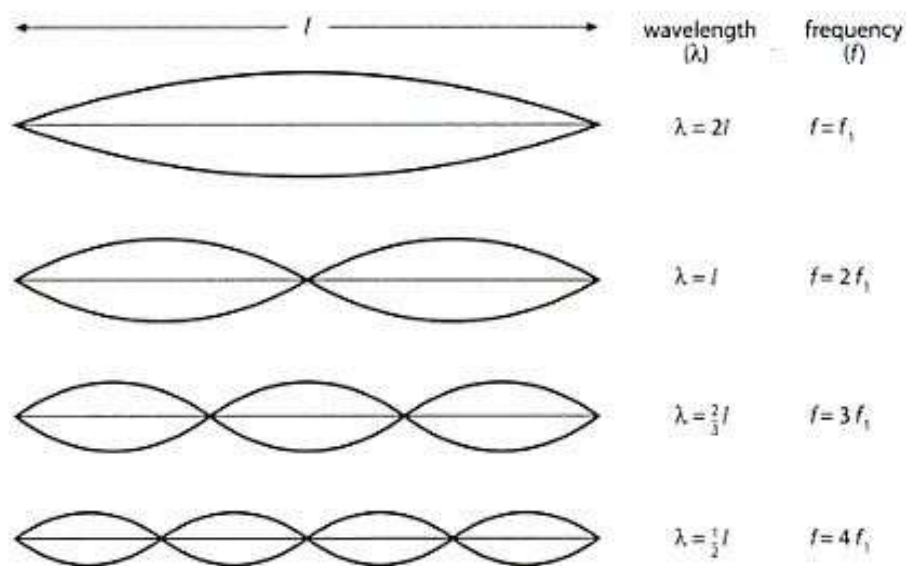
$$E = \mu \omega^2 A^2 L; \quad \mu = \text{Mass} / \text{Length}$$

$$\text{Energy per unit length: } \frac{E}{L} = \mu \omega^2 A^2$$

Example: 1 Clamped string

$$\text{Fundamental frequency / First harmonics: } \frac{\lambda_0}{2} = L \Rightarrow \lambda_0 = 2L$$

$$f_0 = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$



$$\text{First overtone: } f_1 = \frac{2}{2L} \sqrt{\frac{T}{\mu}} = 2f_0$$

$$\text{Second overtone: } f_3 = \frac{3}{2L} \sqrt{\frac{T}{\mu}} = 3f_0$$

$$\text{General formula: } f_p = \frac{p}{2L} \sqrt{\frac{T}{\mu}}$$

Example: 2 Clamped at end

$$\text{Fundamental frequency: } \frac{\lambda_0}{4} = L \Rightarrow \lambda_0 = 4L$$

$$f_0 = \frac{1}{4L} \sqrt{\frac{T}{\mu}}$$

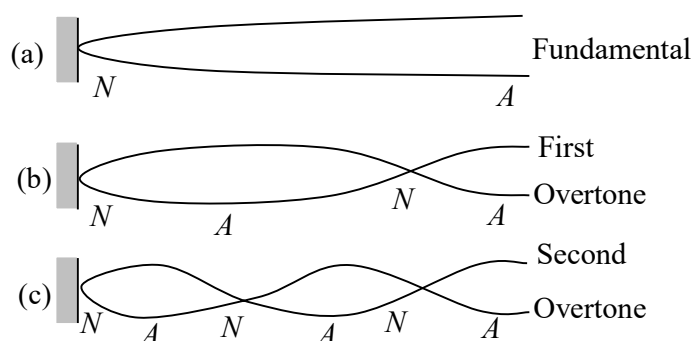
First Overtone: $\frac{3\lambda_1}{4} = L \Rightarrow \lambda_1 = \frac{4}{3}L$

$$f_1 = \frac{3}{4L} \sqrt{\frac{T}{\mu}} = 3f_0$$

Second Overtone

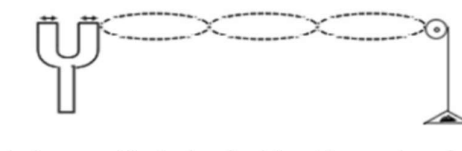
$$\frac{5\lambda_2}{4} = L \Rightarrow \lambda_2 = \frac{4}{5}L$$

$$f_2 = \frac{5}{4L} \sqrt{\frac{T}{\mu}} = 5f_0$$



Melde's experiment

Melde's experimental set up consists of a light string tied to one of the prongs of a tuning fork which is mounting on a sounding board. The other end of the string is passed over a horizontal pulley and a light pan is suspended from the adjusted by changing the weights placed in the pan while the length of the string can be adjusted by changing the position of the pulley towards or away from the fork. If the fork is adjusted so that its arms vibrate parallel to the length of the string. Through adjusting the length or the tension number of clear loops can be visible as the standing wave is formed. When the prongs vibrate parallel to the length of string, the frequency of the fork is twice the frequency of the string. Since the string is vibrating in the fundamental mode, its frequency " n " is given by



$$n = \frac{1}{2L} \sqrt{\frac{T}{\mu}}, \text{ Therefore, the frequency of the tuning fork will be } N = 2n = \frac{1}{L} \sqrt{\frac{T}{\mu}}$$

If number of loops formed is " p ", then $N = \frac{p}{L} \sqrt{\frac{T}{\mu}}$.

Wave Motion

Open organ pipe

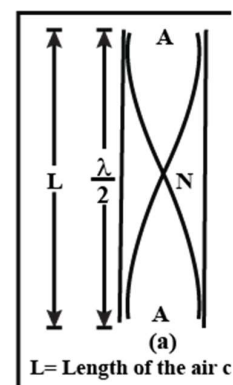
The pipe in which the both of its ends are open is called open organ pipe. Flutes is the example of organ pipe.

1. First mode of vibration/First harmonic/Fundamental frequency

In the mode of vibration in the organ pipe, two antinodes are formed at two open ends and one node is formed in between them. If length of the pipe be 'l' and be the wavelength of wave emitted in this mode of vibration. Then,

$$l = \frac{2\lambda_0}{4} \Rightarrow \lambda_0 = 2l$$

Frequency $f_s = \frac{v_s}{2l}$



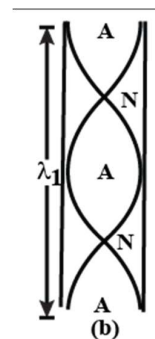
2. Second mode of vibration/First overtone /2nd harmonics

In the second mode of vibration in the open organ pipe, there antinodes are formed at two ends and two nodes between them.

If 'l' be the length of the pipe and λ_1 be the wavelength of wave emitted in this mode of vibration. Then,

$$l = \lambda_1$$

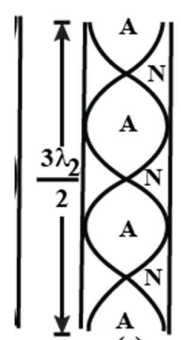
Frequency $f_1 = \frac{v_s}{l} = 2f_0$



3. Second mode of vibration/second overtone/3rd Harmonics

$$l = 3/2\lambda_2$$

Frequency $f_2 = \frac{3v_s}{2l} = 3f_0$



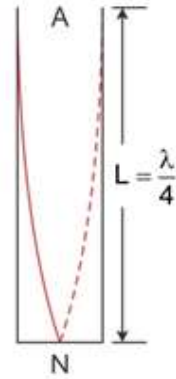
Closed organ pipe

Closed organ pipe is the one in which only one end is open and the other is closed and then sound is passed.

1. Fundamental frequency

$$l = \frac{\lambda_0}{4} \Rightarrow \lambda_0 = 4l$$

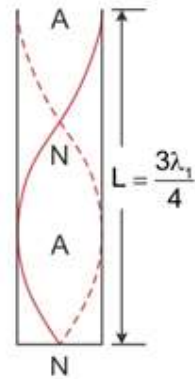
Frequency $f_0 = \frac{v_s}{4l}$ -First harmonics



2. First overtone

$$l = \frac{3\lambda_1}{4} \Rightarrow \lambda_1 = \frac{4l}{3}$$

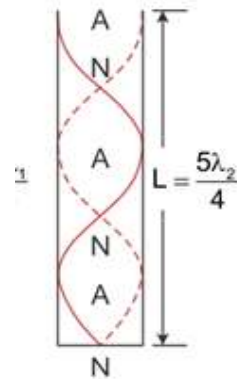
$f_1 = \frac{3v_s}{4l} = 3f_0$ -Second harmonics



3. Second overtone

$$l = \frac{5\lambda_2}{4} \Rightarrow \lambda_2 = \frac{4l}{5}$$

$f_2 = \frac{5v_s}{4l} = 5f_0$ -Third harmonics



Phase Velocity

The phase velocity of a wave is the rate at which the wave propagates in any medium. This is the velocity at which the phase of any one frequency component of the wave travels. For such a component, any given phase of the wave (for example, the crest) will appear to travel at the phase velocity.

We know that the wave equation can be written as

$$y = A \sin(\omega t - kx)$$

The phase velocity is defined as

$$v_p = \frac{\omega}{k}$$

In terms of wave length (λ) and time period (T) the phase velocity can be written as

$$v_p = \frac{\lambda}{T}$$

Group Velocity

The velocity with which the maximum amplitude of the wave group propagates in the medium is known as group velocity.

$$y_1(t) = a \sin(\omega_1 t - k_1 x)$$

$$y_2(t) = a \sin(\omega_2 t - k_2 x)$$

$$y = y_1(t) + y_2(t)$$

$$= 2a \sin\left(\left(\frac{\omega_1 - \omega_2}{2}t\right) - \left(\frac{k_1 - k_2}{2}x\right)\right) \times \cos\left(\left(\frac{\omega_1 + \omega_2}{2}t\right) - \left(\frac{k_1 + k_2}{2}x\right)\right)$$

$$v_g = \frac{\frac{\omega_1 - \omega_2}{2}}{\frac{k_1 - k_2}{2}} = \frac{\omega_1 - \omega_2}{k_1 - k_2} = \frac{d\omega}{dk}$$

Relation between group velocity and phase velocity

$$v_g = \frac{d(kv_p)}{dk} = v_p + k \frac{d(v_p)}{dk}, \quad k = \frac{2\pi}{\lambda} \rightarrow dk = -\frac{2\pi}{\lambda^2} d\lambda$$

$$v_g = v_p - \lambda \frac{d(v_p)}{d\lambda}$$

Case: 1

If $\frac{d(v_p)}{d\lambda} = 0$, non-dispersive medium

$$v_g = v_p$$

Example: $\omega = ck$; $v_p = \frac{\omega}{k} = c$ and $v_g = \frac{d\omega}{dk} = c$

Case: 2

If $\frac{d(v_p)}{d\lambda} = +Ve$ Then it is called Normal dispersive medium

$$v_g < v_p$$

Example: Propagation of em wave in plasma

$$\omega^2 = c^2 k^2 + \omega_p^2; \quad \omega_p = \text{Plasma frequency}; \quad \omega_p = \sqrt{\frac{n_0 e^2}{m \epsilon_0}}$$

$$\frac{\omega^2}{k^2} = c^2 + \frac{\omega_p^2}{k^2}$$

$$\frac{\omega}{k} = v_p = \sqrt{c^2 + \frac{\omega_p^2}{k^2}} > c$$

$$2\omega \frac{d\omega}{dk} = 2c^2 k \Rightarrow v_g = \frac{c^2}{v_p} = \frac{c^2}{\sqrt{c^2 + \frac{\omega_p^2}{k^2}}} < c$$

Case: 3

If $\frac{d(v_p)}{d\lambda} = -Ve$ Then it is called Anomalous dispersive medium

$$v_g > v_p$$

Doppler's effect

If there is relative motion between source of sound and observer, then the apparent frequency heard by the observer will be different from the actual frequency of source. This effect is known as Doppler effect.

Doppler effect formula:

Let us define few parameters

f' = Observed frequency; f = Actual frequency

v = Velocity of sound waves; v_o = Velocity of observer;

v_s = Velocity of source.

$$f' = \frac{(v \pm v_o)}{(v \mp v_s)} f$$

Case: 1

Source moving towards observer at rest

$$f' = \frac{v}{(v - v_o)} f$$

Case: 2

Source moving away from observer at rest

$$f' = \frac{v}{(v + v_o)} f$$

case: 3

Observer moving towards the stationary source

$$f' = \frac{(v + v_o)}{v} f$$

Case: 4

Observer moving away from the stationary source

$$f' = \frac{(v - v_o)}{v} f$$

Case: 5

Both are moving towards each other

$$f' = \frac{(v + v_o)}{(v - v_s)} f$$

Case: 6

Both are moving away from each other

$$f' = \frac{(v - v_o)}{(v + v_s)} f$$

Fermat Principle

Optical path:

When light travels a distance l in a medium of refractive index n the product nl is termed the optical path corresponding to the distance l . By definition,

$$n = \frac{c}{v}$$

Thus, the optical path $nl = \frac{c}{v}l = c \times$ Time taken by light to travel the distance l in the medium.

Fermat's Principle

The path followed by a ray of light in moving from one point to another point after any number of reflections or refractions would always be stationary (i.e., either maximum or minimum) with respect to variations of that path.

If the light from one point reaches another point directly or after reflection or refraction at a plane surface the path is minimum. Hence time taken by light in this case must also be minimum. Thus, the principle of least path may also be called the principle of least time. If the reflection or refraction occurs at spherical surfaces the path may be either maximum or minimum, depending on the curvature of the surfaces. Hence, in general, the path would be stationary.

The optical path between two points may be expressed as $\sum nl$ or $\int n dl$

From Fermat's principle,

$$\delta \sum nl = \delta \int n dl = 0 \dots (1)$$

Rectilinear propagation of light:

If light moves from one point to another in a homogeneous medium, Fermat's principle implies that path of the ray between the two points should be least. Obviously, the path is a straight line. Thus, the rectilinear propagation of light results as direct consequence of Fermat's principle.

Principle of reversibility of light:

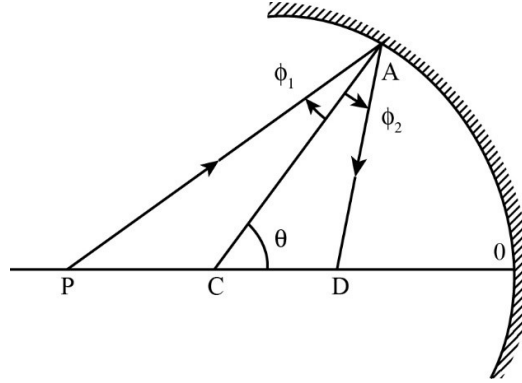
Fermat's principle stated in the form of the Equation (1), involves only the optical path and not the direction of propagation of light. So a ray in going from the point P to Q will trace the same route as one from Q to P . Thus, the principle of reversibility is also included in Fermat's principle.

To establish laws of reflection at a spherical surface from Fermat's principle:

Suppose a ray of light from a fixed point P is arriving at another fixed point Q via the point A of a spherical surface as shown in Figure below. Let ϕ_1 and ϕ_2 be the angles which the rays PA and

AQ make with the normal CA at A drawn from the centre of curvature C of the spherical surface.

Let $OC = R =$ radius of the spherical surface. Now from $\triangle APC$ and $\triangle AQC$ we can write



$$PA = (R^2 + CP^2 + 2R \cdot CP \cdot \cos \theta)^{\frac{1}{2}}$$

$$AQ = (R^2 + CQ^2 - 2R \cdot CQ \cdot \cos \theta)^{\frac{1}{2}} \dots (2)$$

According to Fermat's principle the total optical path $PA + AQ$ between P and Q should be stationary and hence the variation of $PA + AQ$ with the angle θ must be zero. Thus,

$$\frac{d}{d\theta} (PA + AQ) = 0$$

Since R , CP and CQ are constants we get from above Equation

$$-\frac{CP}{PA} + \frac{CQ}{AQ} = 0 \Rightarrow -\frac{\sin \phi_1}{\sin (180^\circ - \theta)} + \frac{\sin \phi_2}{\sin \theta} = 0$$

or, $\phi_1 = \phi_2$ i.e, angle of incidence (ϕ_1) = angle of reflection (ϕ_2)

Reflection formula:

$$\cos \theta = 1 - \frac{\theta^2}{2}$$

Now assuming $OP = x$ and $OQ = y$ as the magnitude of distances we can write from equation

(2)

$$\begin{aligned} PA &= \left[R^2 + (x - R)^2 + 2R(x - R) \left(1 - \frac{\theta^2}{2} \right) \right]^{\frac{1}{2}} \\ &= [x^2 - R(x - R)\theta^2]^{\frac{1}{2}} = [x^2 - R(x - R)\theta^2]^{\frac{1}{2}} \\ &= x - \frac{1}{2}R^2 \left(\frac{1}{R} - \frac{1}{x} \right) \theta^2 \dots (3) \end{aligned}$$

Similarly, we get

$$AQ \simeq y - \frac{1}{2}R^2 \left(\frac{1}{R} - \frac{1}{y} \right) \theta^2 \dots (4)$$

Now for the optical path $L_{op} = PA + AQ$ to an extremum we must have $\frac{dL_{op}}{d\theta} = \frac{d}{d\theta} (PA + AQ) = 0$

Using Equation (3) and (4) we get, $\left[\frac{2}{R} - \frac{1}{x} - \frac{1}{y}\right] \theta = 0 \dots\dots\dots (5)$

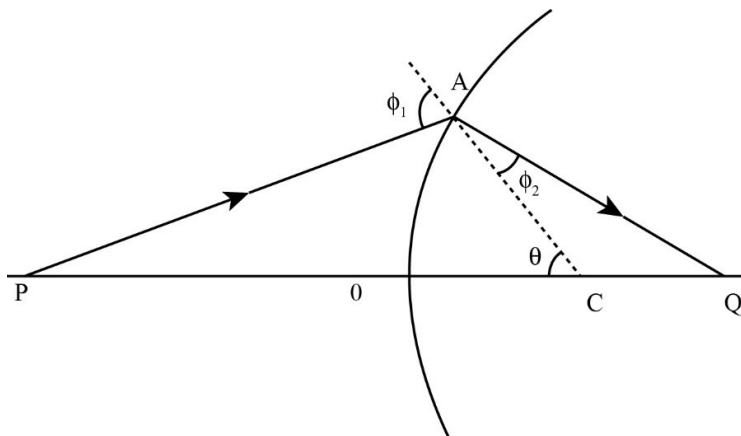
If the quantity within the bracket is not zero then we must have $\theta = 0$. In this case light falls normally on the surface along PO and gets reflected normally back to Q along OQ. Laws of reflection is obeyed here. On the other hand, if $\frac{1}{y} + \frac{1}{x} = \frac{2}{R} \dots\dots (6)$

$\frac{dL_{op}}{d\theta}$ would vanish for all values of θ . This indicates that all paths like PAQ are allowed ray paths i.e., all paraxial rays emanating from P will, after reflection, pass through Q. Therefore, in this case Q will be the image point of the object point P. Thus Equation (6) is reflection formula at the concave spherical surface. Using the sign convention that all the distances measured to the left of O are negative we may write object distance $OP = u = -x$, image distance $OQ = v = -y$ and radius of curvature $OC = r = -R$. Then formula (6) may be rewritten for a concave spherical reflector as

$$\frac{1}{v} + \frac{1}{u} = \frac{2}{r}$$

To establish laws of refraction at a spherical surface from Fermat's principle:

Suppose a ray of light from fixed point Q via the point A on a spherical surface separating two media of refractive indices n_1 and n_2 . Let ϕ_1 and ϕ_2 be the angles which the rays PA and AQ make with the normal CA at A drawn from the centre of curvature C of the spherical surface. From $\triangle APC$ and $\triangle AQC$ we can write



$$PA = (AC^2 + CP^2 - 2 \cdot AC \cdot CP \cdot \cos \theta)^{1/2} \text{ and } AQ = (AC^2 + CQ^2 + 2 \cdot AC \cdot CQ \cdot \cos \theta)^{1/2} \dots\dots(7)$$

According to Fermat's principle the total optical path $n_1 PA + n_2 AQ$ between P and Q should be stationary and hence

$$\frac{d}{d\theta}(n_1 PA + n_2 AQ) = 0 \dots (8)$$

Since AC , CP and CQ are constants, we get from Equation (8). Using

$\Rightarrow n_1 \sin \phi_1 = n_2 \sin \phi_2$ which is the snell's law of refraction.

Refraction formula:

Under paraxial approximation the angle θ as indicated in (Figure above) is small so that we can use

$$\cos \theta = 1 - \frac{\theta^2}{2}$$

Now putting $OP = x$, $OQ = y$ and $AC = OC = R$ as the magnitude of the distances we can write from Equation (7),

$$\begin{aligned} PA &= \left[R^2 + (x + R)^2 - 2R \cdot (x + R) \left(1 - \frac{\theta^2}{2} \right) \right]^{\frac{1}{2}} \\ &= [x^2 + R(x + R)\theta^2]^{\frac{1}{2}} \simeq x \left[1 + \frac{R(x + R)\theta^2}{2x^2} \right] = x + \frac{1}{2}R^2 \left(\frac{1}{x} + \frac{1}{R} \right) \theta^2 \\ AQ &\simeq y + \frac{1}{2}R^2 \left(\frac{1}{R} - \frac{1}{y} \right) \theta^2 \end{aligned}$$

Now for the optical path $L_{op} = n_1 PA + n_2 AQ$ be an extremum we must have

$$\frac{dL_{op}}{d\theta} = \frac{d}{d\theta} [n_1 PA + n_2 AQ] = 0, \Rightarrow \left[\frac{n_2}{y} + \frac{n_1}{x} - \frac{n_2 - n_1}{R} \right] \theta = 0$$

$$\left[\frac{n_2}{y} + \frac{n_1}{x} - \frac{n_2 - n_1}{R} \right] = 0 \dots (9)$$

$\frac{dL_{op}}{d\theta} = 0$ for all values of θ . This indicates that all paths like PAQ are allowed ray paths i.e., all paraxial rays emanating from P will, after refraction, pass through Q. Therefore, Q in that case will be the image point of the object point P. Equation (9) then represents the refraction formula at the curved surface.

Using the sign convention that all distances measured to the left of O are negative and those to its right positive we may write object distance $OP = u = -x$, image distance $OQ = v = y$ and radius of curvature $OC = r = R$. Then formula ... may be rewritten for a convex spherical surface in form

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{r}$$

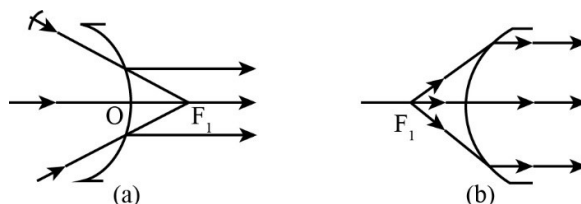
Refraction At Spherical Surfaces

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{r}$$

where u = object distance, v = image distance, r = radius of curvature

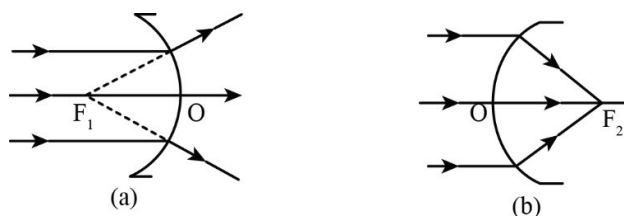
This equation is known as Gaussian formula for a single refracting surface. Here is called the $\frac{n_2 - n_1}{r}$ power of the refracting surface.

Principle Foci of a Spherical Surface



The first focal point (F) is a point on the principal axis such that rays moving towards that point [Figure above] or diverging from that point [Fig. above] become parallel to the principal axis after refraction. The distance of F_1 from the pole of the refracting surface is called the first principal focal length (f) of the surface. Hence $f' = u$ when $v = \infty$. From the general formula (1)

$$f = -\frac{n_1 r}{n_2 - n_1} \dots \dots (2)$$



The second focal point (F_2) is a point on the principal axis such that the incident rays moving parallel to the principal axis will, after refraction, either appear to diverge from this point [Figure above]) or actually converge to this point-[Figure above]]. This distance of F_2 from the pole of the refracting surface is called the second principal focal length (f) of the surface. Hence $f = v$ when $u = \infty$. From the general formula (1).

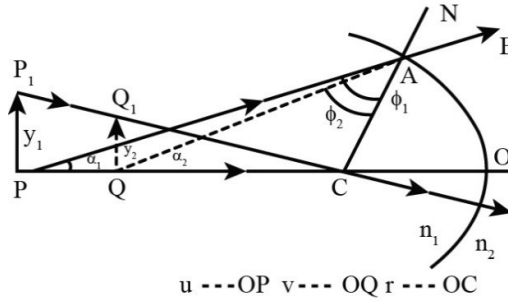
$$f = \frac{n_2 r}{n_2 - n_1} \dots \dots (3)$$

we get,

$$\frac{f}{f'} = -\frac{n_2}{n_1} \dots \dots (4)$$

This indicates that the two principal focal lengths are always of opposite signs.

Magnification for A Spherical surface:



In Figure above let PP1 be an object and QQ1 be its image formed by refraction at the concave spherical surface separating two media of refractive indices n_1 and n_2 .

Lateral or linear magnification:

Linear magnification (m) is defined as the ratio of the image height (y_2) to the object height (y_1) both measured perpendicular to the axis. Thus

$$m = \frac{y_2}{y_1} = \frac{OQ - OC}{OP - OC} \dots (5)$$

Now according to our sign conventional distances measured to the left are negative. Let the object distance $u = -OP$, image distance $v = -OQ$ and the radius of curvature $r = OC$.

$$m = \frac{v-r}{u-r} \dots (6)$$

$$n_1 \left(\frac{1}{r} - \frac{1}{u} \right) = n_2 \left(\frac{1}{r} - \frac{1}{v} \right) \Rightarrow n_1 \frac{u-r}{u} = n_2 \frac{v-r}{v} \dots (7)$$

Hence from Equation (6) and (7), $m = \frac{n_1}{n_2} \cdot \frac{v}{u} \dots (8)$

Longitudinal magnification:

If a small object of length δu , placed along the axis produces an images of length δv , the lim

$\frac{\delta v}{\delta u} = \frac{dv}{du}$ is a measure of longitudinal magnification (l).

Differentiating the equation, $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{r}$ we get,

$$\begin{aligned} -\frac{n_2}{v^2} \cdot \frac{dv}{du} + \frac{n_1}{u^2} &= 0 \Rightarrow \frac{dv}{du} = \frac{n_1}{n_2} \cdot \frac{v^2}{u^2} \\ \Rightarrow l = \frac{dv}{du} &= \left(\frac{n_1 v}{n_2 u} \right)^2 \cdot \frac{n_2}{n_1} = m^2 \cdot \frac{n_2}{n_1} \dots (9) \end{aligned}$$

Angular magnification:

If the incident ray PA and the corresponding refracted ray AB make angles α_1 and α_2 respectively with the axis (Figure above) the ratio $\frac{\tan \alpha_2}{\tan \alpha_1}$ is called the angular magnification (M).

$$\Rightarrow M = \frac{\tan \alpha_2}{\tan \alpha_1} = \frac{AO/OQ}{AO/OP} = \frac{OP}{OQ} = \frac{u}{v} \dots (10)$$

Using Equation (8) we can write, $M = \frac{n_1}{n_2} \cdot \frac{1}{m} \dots (11)$

Inter-relation:

Combining Equation (9) and (11) we get $l = m \times \frac{1}{M}$, $\frac{l \times M}{m} = 1$

$$\Rightarrow \frac{\text{Longitudinal magnification} \times \text{Angular magnification}}{\text{Linear magnification}} = 1 \dots (12)$$

Lagrange's Law and Helmholtz's Relation:

In the paraxial region there exists an important relation between object size, image size and the angles which the conjugate rays make with the principal axis. Let in Figure above PP_1 be a small object of size y_1 placed transversely to the axis in front of a certain refracting surface C is the centre of curvature and QQ_1 is the image of size y_2 . The conjugate rays PA and AB make angles α_1 and α_2 respectively with the axis. From Figure above we can write,

$$\frac{y_2}{y_1} \cdot \frac{\tan \alpha_2}{\tan \alpha_1} = \frac{QQ_1}{PP_1} \times \frac{AO/OQ}{AO/OP} \quad [\text{in the paraxial region the point A is very close to pole O}]$$

According to our sign convention all distances measured to the left of O are negative. Let the object distance $u = -OP$, image distance $v = -OQ$ and radius of curvature $r = -OC$.

Therefore,

$$\frac{y_2}{y_1} \cdot \frac{\tan \alpha_2}{\tan \alpha_1} = \frac{v-r}{u-r} \times \frac{u}{v} \dots (13)$$

Now we have the relation

$$\begin{aligned} \frac{n_2}{v} - \frac{n_1}{u} &= \frac{n_2 - n_1}{r} \Rightarrow n_1 \left(\frac{1}{r} - \frac{1}{u} \right) = n_2 \left(\frac{1}{r} - \frac{1}{v} \right) \\ \Rightarrow \frac{n_1}{n_2} &= \frac{v-r}{u-r} \cdot \frac{u}{v} \dots (14) \end{aligned}$$

Combining Equation (13) and (14), we get

$$n_1 y_1 \tan \alpha_1 = n_2 y_2 \tan \alpha_2$$

In general, $n_1 y_1 \tan \alpha_1 = n_m y_m \tan \alpha_m \dots (15)$

Its importance lies in the fact that it expresses linear magnification in terms of angular magnification. Note that this relationship between the magnifications is independent of the refractive indices of the intervening media.

A system is known as telescopic system if all the cardinal points lie at infinity. In this case, usually $n_i = n_{\dots}$. Hence we get

$$\frac{y_1}{y_m} = \frac{\tan \alpha_m}{\tan \alpha_1} \dots (16)$$

Thus, in a telescopic system angular magnification ($\tan \alpha_m / \tan \alpha_1$) is equal to the reciprocal of linear magnification (y_m / y_1)

ABBE'S Sine Condition:

$$n_1 y_1 \sin \alpha_1 = n_2 y_2 \sin \alpha_2$$

This is known as Abbe's sine condition. It holds good for all rays whether paraxial or not.

When the rays are paraxial i.e α_1 and α_2 are small we can write $\sin \alpha_1 \approx \tan \alpha_1$ and $\sin \alpha_2 \approx \tan \alpha_2$. In this case Abbe's sine condition reduces to Helmholtz's relation.

Refraction Through Thin Lens:

Some important terms:

(i) Principal axis: Principal axis of a lens is the straight line joining the centres of curvature of its two bounding surfaces.

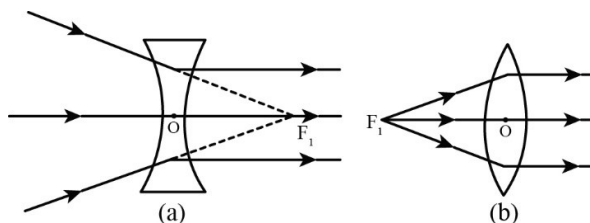
(ii) Poles: The points of intersection of the two surfaces with the principal axis are called the poles of the lens. The distance between the two poles is called the thickness of the lens (which might be considered zero for thin lens),

(iii) Principal Section: The section of lens by a plane through its principal axis is called principal section.

(iv) Aperture: When the bounding line of the lens is circular, the diameter of this circular boundary is called aperture of the lens.

(v) Optical centre: Optical centres of a lens is a point on the principal axis such that any ray through it emerges parallel to its direction of incidence. For equi-convex lens the optical centre is situated at the mid-point.

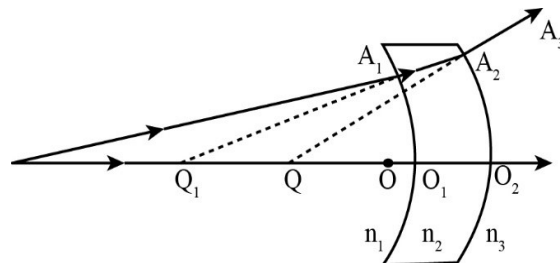
(vi) First focal point: first focal point of a lens is a point on the principal axis such that rays moving towards that point [Figure below] or diverging from that point [Figure below] become parallel to the principal axis after refraction. The distance of F_1 from the optical centre (O) of the lens is called the first focal length (f'), when $v = \infty$.



(vii) Second focal length: Second focal length (F_2) is a point on the principal axis such that the incident rays moving parallel to the principal axis will, after refraction, either appear to diverge from this point [Figure above] or actually converge to point [Figure above]. The distance of F_2 from the optical centre (O) of the lens is called the second focal length (f). Hence $f = v$ when $u = \infty$. According to our sign convention f is negative for a diverging lens and positive for a converging lens.

Refraction At Spherical Surfaces

Lens formula with optical centre as origin:



$$\frac{n_3}{v} - \frac{n_1}{u} = \frac{1}{F} \dots \dots (17)$$

where, u = object distance, v = image distance, $\frac{1}{F} = \frac{n_2 - n_1}{r_1} - \frac{n_2 - n_3}{r_2} \dots \dots (18)$

According to the definition of focal lengths, $u = f'$ (1st focal length) when $v = \infty$;

$v = f$ (2nd focal length) when $u = \infty$;

Putting these conditions in equation (17), we get

$$-\frac{n_1}{f'} = \frac{n_3}{f} = \frac{1}{F} \dots \dots (19)$$

$$\Rightarrow \frac{f'}{f} = -\frac{n_1}{n_3} = -\frac{\text{r.i. of incident medium}}{\text{r.i. of emergent medium}}$$

Combining (17) and (19) we get, $\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_3}{f} \dots \dots (20)$

Case-I: When $n_1 = n_3 \Rightarrow \frac{1}{f} = -\frac{1}{f'} = \left(\frac{n_2}{n_1} - 1\right) \left(\frac{1}{r_1} - \frac{1}{r_2}\right) \dots (21)$

And $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \dots \dots (22)$

Case-II: When the surrounding medium is air, $n_1 = n_3 = 1$ and if $n_2 = n$ then $\frac{1}{f} = \frac{1}{f'} = (n - 1) \left(\frac{1}{r_1} - \frac{1}{r_2}\right)$

Sum of the powers of the refracting surfaces.

The power of the lens is $P = \frac{1}{F} = \frac{n_2 - n_1}{r_1} + \frac{n_3 - n_2}{r_2}$ Sum of the powers of the refracting surfaces.

Conjugate foci relation with the two foci as origin Newton's equation:

The conjugate foci relation for a thin lens with optical centre (O) as origin is given by

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

where we assume that the media on both sides of the lens are same.

$$\frac{f}{v} - \frac{f}{u} = 1 \Rightarrow \frac{f}{v} + \frac{f'}{u} = 1 \Rightarrow uv' - vf' + ff' = ff'$$

$$(u - f')(v - f) = ff'$$

Here U and V represent the object distance and its image distance measured from the first and second foci respectively.

$$UV = ff'$$

This equation is known as Newton's equation.

Equivalent Lens and Equivalent Focal Length:

A single lens is said to be equivalent to a system of lenses when that single lens produces an image of a given object at the same place and of the same size as that formed by the system of lenses. The focal length of such a single lens is called equivalent focal length of the system of lenses. The equivalent focal length F of two thin lenses having second focal lengths f_1 and f_2 and kept separated by a distance t in air is given by ,

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{t}{f_1 f_2}$$

Some Important Problems with Lenses:

Minimum distance between an object and its real image produced by a convex lens:

Suppose we consider a convex lens in air forming real image on its right-hand side for an object placed on the left. Here object distance u is $-ve$, image distance v and focal length f are $+ve$.

So, by putting numerical values of u, v and f in the formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ (1)

For the distance $D = u + v$ to be minimum, $\frac{d}{du}(u + v) = 0 \Rightarrow \frac{dv}{du} = -1$ Now from Equation(1), on

differentiation, $-\frac{1}{v^2} - \frac{1}{u^2} \frac{du}{dv} = 0$

Therefore, for D to be minimum, $-\frac{1}{v^2} - \frac{1}{u^2}(-1) = 0 \Rightarrow u = v$ Thus, from (1) $u = v = 2f$

Thus $D_{\min} = u + v = 2f + 2f = 4f$

Condition for the formation of two real images of a given object, on the same screen, for two different positions of a convex lens and expression for focal length in terms of lens displacement:

Here the distance between the object and the screen i.e., $D = u + v$ is constant. Therefore, putting $v =$

$D - u$ in Equation (1) we get, $\frac{1}{D-u} + \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow u^2 - Du + Df = 0 \dots (2)$$

For real distinct roots of this quadratic equation we must have $D^2 > 4Df \Rightarrow D > 4f$

So, for getting two real images on the screen the distance between the object and the screen must be greater than four times the focal length of the convex lens.

Solving Equation (2) we get for the two positions of the lens

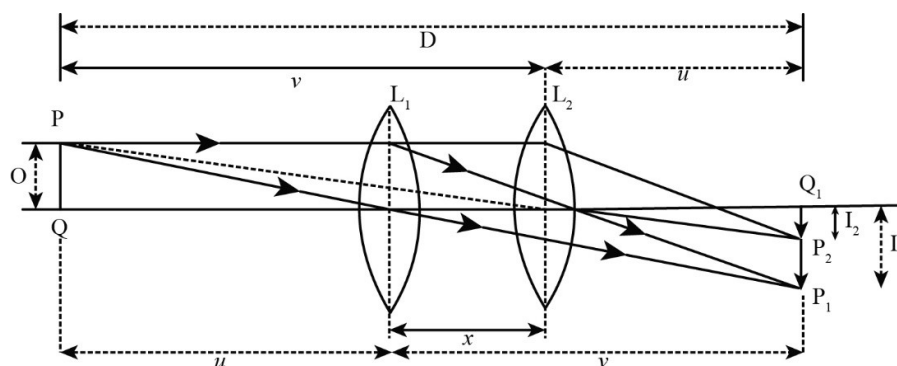
$$u_1 = \frac{D + \sqrt{D^2 - 4Df}}{2} \text{ and } u_2 = \frac{D - \sqrt{D^2 - 4Df}}{2} \dots (3)$$

Therefore, displacement of the lens between the two positions is

$$x = u_1 - u_2 = \sqrt{D^2 - 4Df} \Rightarrow f = \frac{D^2 - x^2}{4D} \dots (4)$$

Size of the object is the geometric mean of the sizes of the images produced by a convex lens at two positions:

Let u, v and I_1 be respectively the object distance, image distance and length of the image for the first position L_1 of the lens; while the corresponding quantities for the second position L_2 of the lens would be v, u and I_2 ($\because$ the positions of object and image are interchangeable) [See Figure below]. If O be the length of the object then magnifications for the two positions are given by



$$m_1 = \frac{I_1}{O} = \frac{v}{u} \text{ and } m_2 = \frac{I_2}{O} = \frac{u}{v}$$

$$\frac{I_1 I_2}{O^2} = 1 \Rightarrow O = \sqrt{I_1 I_2}$$